

Cécile CHAN

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EDUCATION

The Hong Kong Polytechnic University

Hong Kong, HK

BSc (HONS) Scheme in Language Sciences and Analytics

Sept. 2024 - Jun. 2028

- Cumulative GPA: 3.82/4.3 (rank 1) | Latest GPA: 4.04/4.3 | HKSAR Government ROA Scholarship | MITACS Research Scholarship
- Relevant coursework:

Linguistics: Advanced phonetics and phonology, computational linguistics, syntax, psycholinguistics, sociolinguistics, pragmatics

AI and Data Analytics: Statistics (Python, R, and SPSS), data visualization, machine learning

Cognitive Science: Language development and acquisition, neuroimaging techniques (EEG, fMRI, MEG, fNIRS)

Speech-language Pathology: Anatomy and physiology, speech disorders (e.g., CAS, autism), literacy disorder, language disorder assessment for children, operating clinical tools (PRAAT phonetic analysis and Electropalatography)

SKILLS

- **Programming & Statistical Modeling:** Python (pandas, statsmodels, scikit-learn), R (lme4, ggplot2, tidyverse), SPSS (ANOVA, multiple linear regression), MATLAB, Git version control
- **Data Processing & Visualization:**
 - ❖ Acoustic analysis: PRAAT, ASR (automatic speech recognition);
 - ❖ EEG /neuroimaging: Brainstorm, ERPLAB (artifact rejection, filtering);
 - ❖ Linguistic/behavioral: Phonetic transcription (IPA), LanesBox (corpus visualization), Excel;
 - ❖ Visualization: Seaborn, Matplotlib, Plotly
- **Experimental Design & Neuro-imaging:** EEG, fMRI, fNIRS, eye-tracking, electropalatography, acoustic analysis paradigms, survey design (Qualtrics, Google Forms), participant recruitment

RESEARCH EXPERIENCE

Neurocognitive Processing of Tonal Accent Adaptation in Mandarin Speech by Cantonese Listeners

Research Assistant | Neurolinguistics (EEG) PolyU Lab | Supervised by Prof. Eric Pelzl

Sept. 2025 - May. 2026

This EEG (ERP) study investigates how bilingual listeners neurally adapt to systematic tone mispronunciations in accented Mandarin, examining the neurocognitive mechanisms of predictive processing and perceptual adaptation in tonal speech comprehension.

- **EEG Experiment Execution and Data Processing:** Independently conducted the full experiment lifecycle (cap fitting, impedance optimization, task instruction, real-time data quality monitoring), and processed raw neural data (MATLAB and EEGLAB).
- **Stimuli preparation & Model building:** Assisted in stimulus preparation, conducted literature reviews and building on neurocomputational models of speech perception, and contributed to interim research reports and manuscript drafts.

Evaluating effector-independent VR-based and effector-specific tablet-based Chinese writing intervention among non-Chinese speaking (NCS) students in Hong Kong

Researcher | Supervised by Prof. Dustin LAU | HK PolyU Speech Therapy Clinic

Sep. 2025 - Present

This project examines the effectiveness of effector-independent and effector-specific training modes on NCS participants, through a treatment featuring orthographic units of different grain sizes, previously shown to benefit students with written language disorder.

- **Participant Recruitment:** Recruited 45 NCS school-aged children, built rapport, and guided them through intervention sessions.
- **Intervention & Tool Development:** Co-developed VR air-writing and tablet-based drawing games targeting orthographic unit learning and motor representation consolidation.
- **Data Collection & Analysis:** Administered pre/post delayed copying tasks (trained, near-transfer, far-transfer items) and conducted ANOVA analysis.
- **Manuscript Writing:** Contributed to drafting of introduction, methods, and results sections.

Reversal of Emotions in Social Media: A Study of Opposites, Negation, and Irony

Research Assistant | Prof. Sophia Lee & Dr. Cheuk Lam Wong | HK PolyU

Mar. 2025 - Mar. 2026

This project investigates how linguistic cues (negation, opposition, irony) trigger emotion reversals in social media text to improve pragmatic sentiment analysis in LLM.

- **Data Acquisition and Pre-processing:** Engineered a novel large-scale NLP corpus of 50,000+ Weibo posts through web scraping and text-processing (tokenization, feature extraction).
- **Sentiment Analysis:** Implemented syntactic parsing to identify and analyse linguistic cues (e.g., irony, negation, opposite) that predict contextual emotion shifts.

Lexical Prosodic Perception and Production Changes in Bilingual Children with Childhood Apraxia of Speech (CAS)

Research Assistant | Supervised by Prof. Min Ney Wong | HK PolyU Speech Therapy Clinic

Feb. 2025 - Dec. 2025

This project investigates the efficacy of Dynamic Temporal and Tactile Cueing (DTTC) on prosodic perception-production skills in bilingual children with Childhood Apraxia of Speech (CAS) to address critical intervention gaps for tonal-language speakers.

- **Participant Recruitment:** Recruited and consented 40 children and their caregivers, and built rapport.
- **Child-friendly Assessment:** Designed and administered standardized child-friendly baseline assessments for experimental (DTTC) and control groups, and performed behavioral observations to ensure data quality and participant engagement.
- **Multi-faceted Data Analysis:** Quantified prosodic perception by performing phonetic transcription, acoustic analysis (PRAAT), and evaluated pre-/post-treatment data by repeated ANOVA and linear mixed-effects models (R lme4 & Python statsmodels).
- **Literature Review Writing:** Conducted a comprehensive literature review on DTTC efficacy, bilingual assessment paradigms, and neurocomputational models of speech development.

Measuring Unidirectional Intelligibility of Taiwan Mandarin Verb Innovations from a Mainland Mandarin Perspective

Team Leader | Supervised by Prof. Yu Yin Hsu | HK PolyU

Feb. 2025 - May. 2025

This project empirically measured the unidirectional intelligibility of novel verb forms in Taiwan Mandarin among Mainland Mandarin listeners, exploring the impact of linguistic innovation on cross-dialectal communication.

- **Full Data Collection Cycle Management:** Recruited and consented 12 native Mainland Mandarin speakers, conducted auditory perception experiments, cleaned qualitative feedback, resulting in a precise 57.5% intelligibility (Python).
- **Original Weighted Scoring Algorithm Development:** Innovatively devised and applied an original weighted scoring algorithm based on morphemic analysis to quantify intelligibility, moving beyond simple binary (understood/not understood) metrics.

EXTRACURRICULAR ACTIVITIES

Hans Andersen Club 「安徒生會」 Volunteer

Volunteer | Jan. 2026 - Present

- **Reading Ability Intervention Protocol Design:** Collect feedback, observe child engagement, and refine program design to ensure responsiveness to children's evolving needs.
- **Storybook Design:** Designed and developed age-appropriate picture books for children in HK Evangel Children's home.
- **Storytelling Camp Coordination:** Organized group role-playing activities and expression workshops.

Culture Promotion and Events Office (CPEO), PolyU

Student Assistant | Sept. 2025 - Present

- **Event Execution:** Organized and coordinated events (e.g., Annual Mid-autumn Festival Gala) and drama performances.
- **Public Relations & Hosting:** Served as a Master of Ceremonies (MC) for major events; guided guests and managed inquiries at information desks, enhancing visitor experience.
- **Advertising Promotion:** Drafted and disseminated posters (Adobe Creative Suite) & engaging event notification emails.

Knowledge Transfer and Entrepreneurship Office (KTEO), PolyU

Student Ambassador | Sept. 2025 - Present

- **Activity Planning:** Assist in activities and events preparation and execution (e.g. orientation info day, competition, seminars)
- **Administrative Support & InnoHub Operations:** Assist in administrative works and daily operation of PolyU InnoHub.

HONORS

- Student Representative of Student Development and Affairs Committee, PolyU
- Undergraduate Student Representative to the Board of Faculty of Humanities [FB(FH)], PolyU
- Undergraduate Student Representative to the Faculty Learning & Teaching Committee